Project Title: EV Analytics & Visualization Platform **Project Design Phase-I - Solution Fit Template** **Team ID**

Define CS, fit into CC

1. CUSTOMER SEGMENTS

- **Aarav**: Urban Energy/City Planner focused on infrastructure and grid stability.
- **Meera:** Fleet Manager for EV taxi services focused on operational efficiency.
- **Ravi:** Eco-conscious individual consumer looking for data-driven purchasing insights.

CC

6. CUSTOMER CONSTRAINTS

- **Budget:** Limited funds for high-end proprietary analytics software.
- **Data Access:** Difficulty in obtaining real-time data from various vehicle sensors or power grids.
- **Technical Skill:** Users may need insights without having to write complex SQL or code.

5. AVAILABLE SOLUTIONS

- **Manual Tracking**: Using spreadsheets or logbooks to track vehicle mileage and charging.
- **Manufacturer Apps:** Standard apps provided by car brands (often lack cross-brand comparison).
- **Pros/Cons**: Existing tools are often fragmented; they provide data but lack the integrated visualization needed for high-level decision-making.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

J&P

2. JOBS-TO-BE-DONE / PROBLEMS

Infrastructure Planning: Determining where and when to install new charging stations based on demand.

- Operational Reliability: Monitoring battery health to prevent vehicle downtime and unexpected costs.
- Informed Purchasing: Comparing complex variables (range vs. price vs. long-term savings) across different EV models.

RC

9. PROBLEM ROOT CAUSE

- The transition from Internal Combustion Engines (ICE) to EVs creates a "data gap" where hardware (the car) is advancing faster than the software needed to manage the ecosystem (the grid and fleet).

7. BEHAVIOUR

- **Direct:** Checking battery "state of charge" (SoC) and searching for nearby charging ports.
- **Indirect:** Engaging in sustainable city initiatives or researching environmental policy changes

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS

Aarav: Noticing local grid strain or increased EV registrations in specific neighborhoods.

Meera: Frequent complaints about taxi range or rising battery replacement costs.

Ravi: Rising fuel prices or a personal commitment to reducing carbon footprint.

4. EMOTIONS BEFORE AND AFTER

- **Before:** Uncertain about grid load, anxious about vehicle downtime, or overwhelmed by technical EV specs.
- **After:** Confident in infrastructure placement, in control of fleet maintenance, and empowered to make a sustainable purchase

10. YOUR SOLUTION

- **Tableau-Based EV Analytics Dashboard:** A comprehensive visualization tool that integrates charging patterns, battery health metrics, and model efficiency comparisons.
- **Key Features:** Heatmaps for grid demand, predictive battery maintenance alerts, and interactive cost-per-kilometer comparison filters.

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8. CHANNELS OF BEHAVIOUR

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Identify strong TR & EM